

GENERAL SPECIFICATION G7.11

Plant Closed Circuit Television System (Plant CCTV)

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G7.11 PLANT CLOSED CIRCUIT TELEVISION SYSTEM (PLANT CCTV)

G7.11.1 Introduction

- (a) Contractor to include for all work to comprise the whole of the labour, including design, and all materials to form a complete plant CCTV system installation.
- (b) Contractor to include provision to carry out all tests, adjustments and commissioning as is required to give an effective operational installation.
- (c) Contractor to include for all Witness Testing with the Client, or his Representative, and training of the Client's Staff in the pricing.

G7.11.2 Standards and Codes

- (a) Unless otherwise agreed by the S.O., the works are to comply with the relevant requirements of the Standards and Codes listed below or referenced in the body of the specification. Materials or products manufactured in accordance with alternative Standards and Codes may be proposed for approval by the S.O., provided it can be demonstrated that the materials or products comply with the requirements of the specified standards. All Standards and Codes quoted are current version, unless specific year references are noted.
- (b) Provide and install equipment and components in accordance with the following standards:
 - (i) SS CP5;
 - (ii) EN 50132;
 - (iii) Install fibre optic cables in accordance with the requirements and recommendations of BS EN 50174-1, BS EN 50174-2 and BS EN ISO 9000; and
 - (iv) ISO 9000
 - 1) Provide supporting documentation in the form of Quality Assurance System Certificates.

G7.11.3 General

- (a) CCTV system shall be used for picture taking, recording, reproduction, amplification, transmission and distribution of video signals to dedicated receivers or monitors. A CCTV system shall comprise of items for monitoring, transmission and controlling purposes.
- (b) Outdoor cameras shall be not affected by lightning surge current or direct heat of lightning. The Contractor shall provide lightning protection system for each outdoor camera and mounting pole.
- (c) The Contractor shall, as a minimum, furnish and install all CCTV cameras, mounts, housings, power supply systems, UTP cable, fibre optic cable, network cables, network switches, balun connectors, equipment racks, monitors and all other necessary hardware and software for functioning of the complete system.
- (d) CCTV system shall be Internet Protocol (IP) network based digital video system using IP cameras.
- (e) Standard UTP cable, Ethernet-based network cables, fibres and switches shall be utilized for transmission medium and in the distribution network.

- (f) IP-based video encoders shall be utilized to convert any analogue signal to digital video signals.
- (g) High performance servers shall be provided for system management and recording of the digital video.
- (h) Workstation(s) and LCD monitors shall be provided for control and monitoring purposes.
- (i) The CCTV shall be integrated across all plant areas as described in the Particular Specification.
- (j) The network structure shall be scalable and modular to cater for future expansion or reduction of the system without having to shift the point of connection or interrupt the system operation. The CCTV system network shall be linked up via cabling as specified and shall operate as a complete integrated system.
- (k) The system network and the equipment shall be resilient to provide high availability and to facilitate system maintenance activities.
- (l) CCTV system shall be provided with software and hardware security protection against unauthorized modification to the system configuration, setting, viewing and reviewing of any recorded data.
- (m) Viewing of the real-time video and reviewing of the recorded video or both shall not interrupt and affect the capability and performance of the system.
- (n) The hardware equipment offered shall be state-of the-art, standardized commercial off-the-shelf and modular equipment.
- (o) The CCTV network communication shall employ an open system protocol.

G7.11.4 System Architecture and Components

- (a) A CCTV system shall typically consist of:
 - (i) Picture generation equipment, generally in the form of colour cameras;
 - (ii) A video signal transmission and distribution network connected by fibre optic cable;
 - (iii) A central observation suite and other auxiliary observation stations, where the television signals are displayed on monitors;
 - (iv) Other ancillary equipment e.g. mixing, splitting or multiplexing equipment to obtain special display effects on monitors;
 - (v) Recording and playback;
 - (vi) All Pan-Tilt-Zoom (PTZ) cameras shall be provided with an option to allow the user to -pre-determine the locations that he/ she wants to monitor (i.e. preset locations), including the setting of a routine pattern, where necessary. The CCTV camera system shall have a 'default settings' function, which allows the PTZ cameras to auto reset to their original position after pre-determined time duration; and
 - (vii) CCTV System Integrator or vendor shall propose standard codec to achieve optimal compression ratios while ensuring no or little loss of image/ video quality.

G7.11.5 Pan-Tilt-Zoom (PTZ) Network Camera Minimum Requirements

- (a) The network camera shall support open API, standard protocol such as ONVIF (Open Network Video Interface Forum), PSIA (Physical Security Interoperability Alliance), third party Software Development Kit (SDK) and ensuring interoperability between network video products regardless of the manufacturer.
- (b) Image Sensor: 1/1.9" Progressive Scan Megapixel CMOS sensor.
- (c) Resolution: HD 1080p with 1920x1080 effective Pixels or its equivalent, 2 Megapixels.
- (d) Focal Length: 4.5mm~135mm.
- (e) Angle of View: 67°~2.4°.
- (f) Optical Zoom: 30x.
- (g) Focus Control: Auto/Manual.
- (h) Video Compression: H.265/H.264/MJPEG.
- (i) Video Streaming capability: 3 Streams.
- (j) Video Streaming Resolution: 1080P/720P.
- (k) Streaming Methods: Unicast/ Multicast.
- (l) Day/Night: True Day/Night (ICR), colour/ black and white.
- (m) Noise Reduction: Digital Noise Reduction (DNR) (2D/3D).
- (n) Audio Compression: Pulse Code Modulation (PCM) (G.711a, G.711u).
- (o) Interoperability: PSIA, ONVIF, Computer Generated Imagery (CGI).
- (p) Ethernet: RJ45 (10/100Based_T).
- (q) Protocol: IPv4/IPv6, HTTP, HTTPS, SSL, TCP/IP, UDP, UPnP, ICMP, IGMP, SNMP, RTSP, RTP, SMTP, NTP, DHCP, DNS, PPPOE, DDNS, FTP, IP Filter, QoS, Bonjour, 802.1x.
- (r) Power Supply: AC23V/1.5A or Power Over Ethernet (PoE) + (802.3at).
- (s) Power Consumption: 13W, 23W or PoE.
- (t) Horizontal Adjust Angle: 0°~360°; Vertical Adjust Angle: 0°~90°; Auto Flip Angle: 180°.
- (u) Manual Speed: 0.1°-350°/ second for Pan, 0.1°-250°/ second for Tilt.
- (v) Preset speed: 500°/ second for Pan, 500°/ second for Tilt.
- (w) Operating Temperature: -40°C to +60°C.
- (x) Storage Temperature: -20°C to +70°C.
- (y) Ingress Protection: IP67.
- (z) Vandal Resistance: IK10.
- (aa) Humidity: 0% to 90% RH, non-condensing.
- (bb) Anticorrosion standard: WF2, NEMA 4X, C5-M.

G7.11.6 Explosion Proof PTZ Network Camera Minimum Requirements

- (a) The Network Camera shall support open API, standard protocol such as ONVIF (Open Network Video Interface Forum), PSIA (Physical Security Interoperability Alliance), third party Software Development Kit (SDK) and ensuring interoperability between network video products regardless of the manufacturer.
- (b) Explosion Proof PTZ network camera shall be subjected and certified to rigorous dust and immersion test (IP67). Explosion Proof PTZ network camera shall have IP67 ingress protection rating and are made of 316L stainless steel, corrosion-resistant and suitable for use in hazardous area.
- (c) Image Sensor: 1/2.8" Progressive Scan Megapixel CMOS sensor.
- (d) Resolution: HD 1080p with 1920x1080 effective Pixels: 1920x1080 or its equivalent, 2 Megapixels.
- (e) Focal Length: 4.5mm~135mm.
- (f) Angel of View: 67°~2.4°.
- (g) Optical Zoom: 30x.
- (h) Focus Control: Auto/Manual.
- (i) Video Compression: H.265/H.264/MJPEG.
- (j) Video Streaming capability: 3 Streams.
- (k) Video Streaming Resolution: 1080P/720P.
- (l) Streaming Methods: Unicast/ Multicast.
- (m) Day/Night: True Day/Night (ICR), colour/ black and white.
- (n) Noise Reduction: Digital Noise Reduction (DNR) (2D/3D).
- (o) Audio Compression: Pulse Code Modulation (PCM) (G.711a, G.711u).
- (p) Interoperability: PSIA, ONVIF, Computer Generated Imagery (CGI).
- (q) Ethernet: RJ45 (10/100Based_T).
- (r) Protocol: IPv4/IPv6, HTTP, HTTPS, SSL, TCP/IP, UDP, UPnP, ICMP, IGMP, SNMP, RTSP, RTP, SMTP, NTP, DHCP, DNS, PPPOE, DDNS, FTP, IP Filter, QoS, Bonjour, 802.1x.
- (s) Power Supply: AC220V (AC100V ~240V) or Power Over Ethernet (PoE) + (802.3at).
- (t) Power Consumption: 40W, 75W (IR on, Heater On) or PoE.
- (u) Horizontal Adjust Angle: 0°~360°; Vertical Adjust Angle: -90°~90°.
- (v) Manual Speed: 0.1°~40°/ second for Pan, 0.1°~40°/ second for Tilt.
- (w) Preset speed: 40°/ second for Pan, 40°/ second for Tilt.
- (x) Operating Temperature: -40°C to +60°C.
- (y) Storage Temperature: -20°C to +70°C.
- (z) Ingress Protection: IP67.
- (aa) Enclosure Material: 316L Stainless Steel.
- (bb) Humidity: 0% to 90% RH, non-condensing.

- (cc) Design and construction of the equipment or protective system shall comply with Hazardous Area equipment requirements of General Specification G5 – Electrical Works.

G7.11.7 Fixed Type CCTV Camera Minimum Requirements

- (a) Fixed type camera for indoor/ outdoor environment.
- (b) Resolution: HD 1080p with 1920x1080 effective Pixels: 1920x1080 or its equivalent.
- (c) Video compression: H.265/H.264/MJPEG.
- (d) Day/Night: True Day/Night (ICR), colour/ black and white.
- (e) Interoperability: PSIA, ONVIF, Computer Generated Imagery (CGI).
- (f) Ethernet: RJ45 (10/100BaseT).
- (g) Protocol: IPv4/IPv6, HTTP, HTTPS, SSL, TCP/IP, UDP, UPnP, ICMP, IGMP, SNMP, RTSP, RTP, SMTP, NTP, DHCP, DNS, PPPOE, DDNS, FTP, IP Filter, QoS, Bonjour, 802.1x
- (h) Power Supply: AC23V/1.5A or Power Over Ethernet (PoE) + (802.3at).
- (i) Power Consumption: 13W, 23W or PoE.
- (j) Operating Temperature: -40°C to +60°C.
- (k) Storage Temperature: -20°C to +70°C.
- (l) Ingress Protection: IP67.
- (m) Vandal Resistance: IK10.
- (n) Humidity: 0% to 90% RH, non-condensing.
- (o) Anticorrosion standard: WF2, NEMA 4X, C5-M.

G7.11.8 Video Management and Network Recording Server Minimum Requirements

- (a) The video management and recording server shall serve as a platform, providing management of the CCTV system configuration, monitoring of overall status of all CCTV system components and reporting of equipment failures.
- (b) Shall be capable of continuous, scheduled, alarm/ event, and motion-triggered recording, pre- and post-alarm recording, and fully programmable per-channel basis.
- (c) The storage capacity shall be expandable, maximize storage efficiency and offer a time-and-priority-based system that identifies data to be removed when storage time expires.
- (d) Shall offer plug-and-play configuration and data authentication facilities.
- (e) Shall be fully compatible with IP-based CCTV system architecture and have full over-the-network remote control and administration, system diagnostics and error logging.

G7.11.9 System Performance

- (a) Image Quality
 - (i) The images shall also be clear even when camera is in full operation including when the camera is being tilted, panned, or zoomed.
 - (ii) The recorded images shall be, at all times, be accurate, sharply defined and with accurate colour reproduction under normal lighting. For reduced lighting (10 lux) and emergency lighting conditions (10 lux), the recorded images shall minimally be accurate and defined reproduction of the scene in monochrome.
 - (iii) All monitors shall be capable to display colour images and shall possess appropriate adjustment controls (such as contrast, brightness, sharpness and colour).
 - (iv) The displayed picture on monitors shall, at all times, be sharply defined, stable, with accurate colour reproduction, and shall be free of noise, interference, ghosting and pulsing effects. Aspect ratio of the displayed picture shall be maintained with respect to the video source.
 - (v) Sufficient lighting shall be provided for the camera views such that the basic level of lighting shall allow camera view identification of a human face from a distance of about 10m.
- (b) Camera Tamper Protection/ Detection
 - (i) Camera shall be installed in such a way that it is difficult for unauthorised personnel to change the field of view of the camera. This should be achieved by installing the camera at a suitable location/ height (minimum height of 2m from floor level).
- (c) Storage Capacity
 - (i) Sufficient image recording capacity shall be provided to enable both continuous 24-hour recording of the CCTV system and archival of at least one full set of 60 days recording with an additional minimum buffer of 10%.
 - (ii) In the event of hard disk storage failure, the system shall be able to support minimum of RAID 6 array. When the bad disk is replaced by a new one, the array shall be built while the system continues to operate normally.
- (d) Basic Metadata
 - (i) The image recording equipment shall automatically record each camera ID of the camera being recorded and the date and time of the recording (synchronised to Global Positioning System (GPS) time). This information shall always be displayed where it is least likely to obscure or interfere with the image of the main subject.
- (e) Video Recording Playback
 - (i) The recording equipment shall have the features to include the capabilities to replay and normal play, still field, fast forward, rewind, record, stepping frame, visual search – forward and reverse, speed search and stop. It shall also allow fast search by date/ time slider and alarm.

- (ii) The video footage shall be suitable for immediate playback on media player software bundled within common computer operating system (e.g. Windows Media Player and Microsoft) and/ or other open-source media player software.
 - (iii) The system shall have duplex capability or greater so as to allow simultaneous image recording, image report and playback. The system shall be designed to enable the playback of footage without causing interruption to the recording process.
 - (iv) The CCTV system shall provide for the playback of any image from any camera recorded up to 30 days previous and in a controlled environment.
 - (v) During playback, the system shall also allow variable time-control for image selection.
- (f) Image Export
 - (i) The CCTV system shall have the ability to export any image from any camera up to previous 30 days.
 - (ii) The image exported shall have no loss of individual frame quality, change of frame rate. There shall be no duplication or loss of frames in the export process. The system shall not apply any format conversion or further compression to the exported images, as this can reduce the usefulness of the content.
 - (iii) Any original metadata and/ or authentication signatures shall be exported with the images.
- (g) Replay of Exported Images
 - (i) CCTV System Integrator shall provide viewer, a software to allow playback of the image/ video recordings made on the CCTV system. The viewer software shall also be equipped with the capability to export recordings to open-source container and/ or common multi-media container formats such as *.avi (Microsoft), *.mov (Apple QuickTime) and *.mp4 (MPEG) to facilitate investigation work. It shall also be capable of exporting still images in *.jpeg (JPEG File Interchange Format).

G7.11.10 Other Requirements

- (a) All programs and software used shall be licensed. The Contractor shall provide the license, any authentication key that may come with the software and all manuals associated with the program and software at the end of the Contract.
- (b) CCTV system provided shall be protected against lightning induced over voltages in accordance with local codes. All metal frames of any security or CCTV consoles shall be effectively earthed and incoming power supply shall be protected by surge arrestors.
- (c) All equipment, except portable equipment shall be held firmly in place. This shall include cameras, mounts, cables, etc. Fastenings and supports shall be adequate to support their loads.
- (d) Care shall be taken in wiring to avoid damage to the cables and to the equipment. All wiring shall be executed in strict compliance with the local codes. Cables for contact switch and access card level circuits and power circuits shall run in separate conduits.

- (e) Cables shall, as far as possible, be hooked up to equipment components via appropriate ring type cable lugs and screws or insulation displacement type terminal blocks. Where component leads/wires need to be terminated electrically and mechanically, a wrap of turn around a terminal is considered minimum. Sufficient slack shall be allowed at connections to compensate for temperature changes and vibration. No wire shall be taut.
- (f) The Contractor shall test the system to show that its performance satisfies the requirement of this specification. All test equipment shall be professional and supplied by the Contractor.
- (g) Detailed test procedures shall be provided for commissioning. Training of operations personnel shall be included in the scope of the vendor.

G7.11.11 Centralised Plant CCTV Server and Monitoring Equipment

- (a) A centralised database server and system software shall be provided to integrate and manage the Plant CCTV system.
- (b) CCTV software and centralised database shall be based on open platform communication for access to viewing and controlling of all cameras via a standard web browser. The acceptable lag time for real-time feed (0.5 seconds on an average and 1 second during worst case scenario).
- (c) Workstation(s) shall be provide access to the plant CCTV System.

END OF SECTION